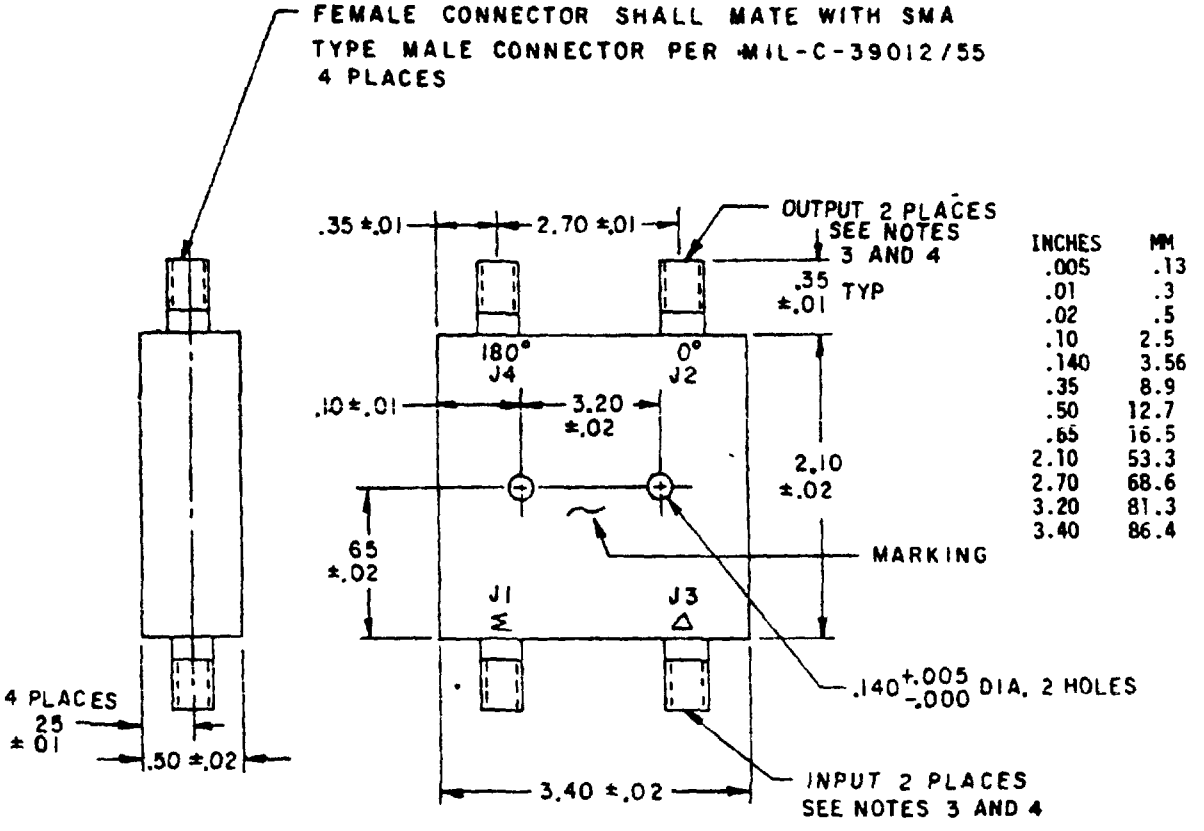
MILITARY SPECIFICATION SHEET

# POWER DIVIDER/COMBINERS, 2-WAY, 180 DEGREES, SMA CONNECTORS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the power divider/combiners described herein shall consist of this document and the latest issue of specification MIL-P-23971.




| INCHES | MM |
|---|---|
| .005 | .13 |
| .01 | .3 |
| .02 | .5 |
| .10 | 2.5 |
| .140 | 3.56 |
| .35 | 8.9 |
| .50 | 12.7 |
| .65 | 16.5 |
| 2.10 | 53.3 |
| 2.70 | 68.6 |
| 3.20 | 81.3 |
| 3.40 | 86.4 |

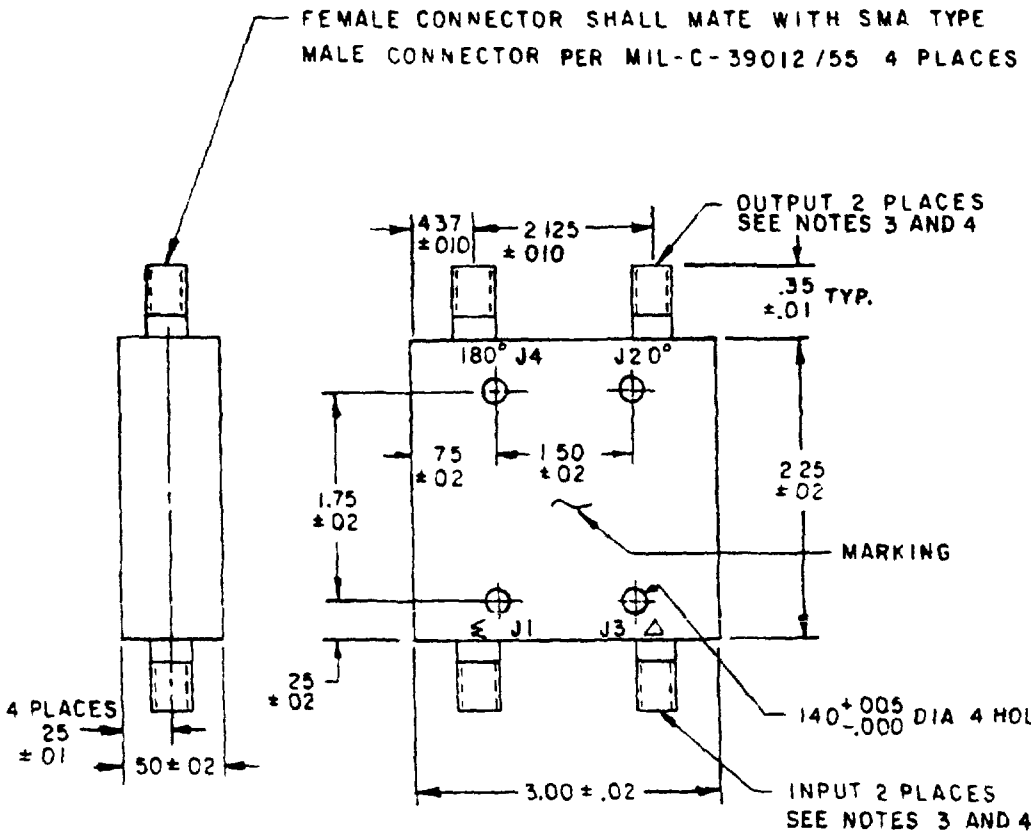
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. When power is applied to the sum Σ (J1) port, the power is divided equally between the 180° (J4) and the 0° (J2) ports, and the power at the output ports J2 and J4 are both in phase with port J1
4. When power is applied to the difference Δ (J3) port, the power is divided equally between the 180° (J4) and the 0° (J2) ports, and the power at the output ports J2 and J4 are both 180° out of phase with port J3.

FIGURE 1 Dimensions and configuration, dash number 01



FSC 5985

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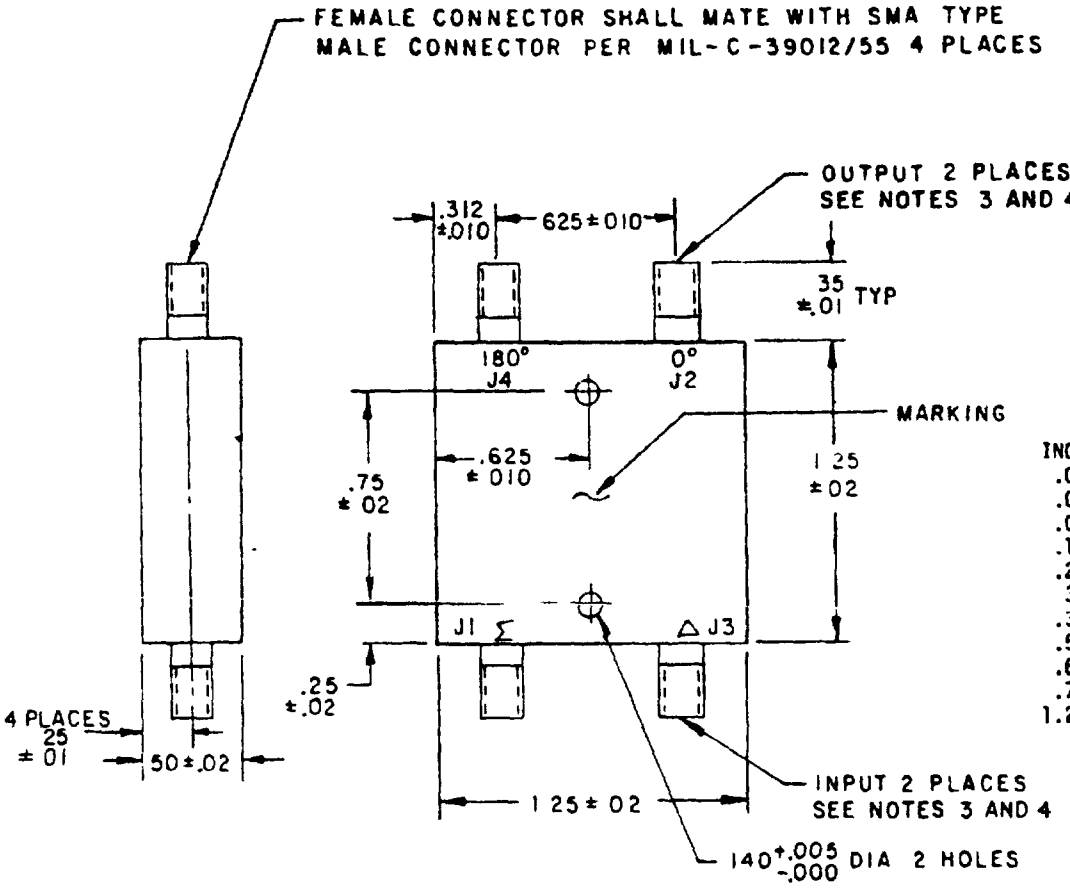

| INCHES | MM |
|---|---|
| .005 | 13 |
| .01 | .3 |
| .02 | .5 |
| .140 | 3.6 |
| .25 | 6.4 |
| .35 | 8.9 |
| .437 | 11.10 |
| .50 | 12.7 |
| .75 | 19.0 |
| 1.50 | 38.1 |
| 1.75 | 44.4 |
| 2.125 | 53.98 |
| 2.25 | 57.2 |
| 3.00 | 76.2 |

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm
3. When power is applied to the sum Σ (J1) port, the power is divided equally between the 180° (J4) and the 0° (J2) ports, and the power at the output ports J2 and J4 are both in phase with port J1.
4. When power is applied to the difference Δ (J3 port, the power is divided equally between the 180° (J4) and the 0° (J2) ports, and the power at the output ports J2 and J4 are both 180° out of phase with port J3.

FIGURE 2 Dimensions and configuration, dash number 02.



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| INCHES | MM |
|---|---|
| .005 | .13 |
| .01 | .3 |
| .02 | .5 |
| .140 | 3.56 |
| .25 | 6.4 |
| 312 | 7.92 |
| .35 | 8.9 |
| .50 | 12.7 |
| .625 | 15.88 |
| .75 | 19.0 |
| 1.25 | 31.8 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. When power is applied to the sum Σ (J1) port, the power is divided equally between the 180° (J4) and the 0° (J2) ports, and the power at the output ports J2 and J4 are both in phase with port J1.
4. When power is applied to the difference Δ (J3 port, the power is divided equally between the 180° (J4) and the 0° (J2) ports, and the power at the output ports J2 and J4 are both 180° out of phase with port J3.

FIGURE 3 Dimensions and configuration, dash number 03.



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TABLE I Electrical performance characteristics and physical requirements

| M23971/13- | Impedance (ohms) | Frequency range GHz | Average coupling (dB) | VSWR max | Insertion loss max (dB) | Isolation min (dB) | Phase balance max (degree) | Amplitude balance (dB) max | Power level | | Weight pounds (grams) | Ambient temperature | | Figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | avg (W) | pk (W) | | Operating | Storage | |
| 01 | 50 | 1-2 | 3 + 2 − 0 | 1.35:1 | 0.5 | 20 | ±6 | ±0.2 | 50 | 4K | 357 (162) | −54° to +55°C | −62° to +85°C | 1 |
| 02 | 50 | 2-4 | 3 + 2 −.0 | 1.5:1 | 0.65 | 20 | ±6 | ±0.4 | 50 | 4K | 344 (156) | −54° to +55°C | −62° to +85°C | 2 |
| 03 | 50 | 4-8 | 3 + 2 − 0 | 1.6:1 | 0.6 | 15 | ±8 | ±0.5 | 50 | 4K | 081 (37) | −54° to +55°C | −62° to +85°C | 3 |

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REQUIREMENTS

Design and construction See figure 1 through 3.

Case. O ring seal

Electrical characteristics See table I

Weight: See tabel I.

Ambient temperature See table I

Environmental tests, In accordance with MIL-P-23971 except

Explosion Applicable

Part number M23971/13 – (dash number from table I)

NOTE Power dividers covered by this specification sheet are substituted for the following manufacturer's part number This information in no way implies that the manufacturer's part is suitable as a subsutute for the military part number

| Dash number | Manufacturer's code | Manufacturer's part number | FSN |
|---|---|---|---|
| 01 | 32562<br>31597 | USCD – 10116 – 1<br>3A0055 3 | T<br>↓ |
| 02 | 32562<br>31597 | USCD – 10116 – 2<br>3A0056-3 | T<br>↓ |
| 03 | 32562<br>31597 | USCD – 10116 – 3<br>30057-3 | 5985-00-459-2153<br>↓ |

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force -
DLA - ES

User activities:
Army -
Navy - AS, OS, MC
Air Force -

Preparing activity
Navy - EC

Agent:
DLA - ES

(Project 5985-0899)



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| | |

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|---|---|
| | ☐ VENDOR |
| b ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
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